IEEE Access
Multidisciplinary | Rapid Review | Open Access Journal

Received 9 February 2025, accepted 17 April 2025, date of publication 23 April 2025, date of current version 2 May 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3563785

RESEARCH ARTICLE

Comprehensive Evaluation of the Impacts of Mixed Traffic Conditions on Urban Networks

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ABSTRACT The emergence of connected and autonomous vehicles (CAVs) has become a focal point in the literature. This study proposes a comprehensive evaluation framework integrating multi-criteria analysis (MCA) methods with traffic microsimulation modeling to assess the impacts of mixed traffic conditions, comprising CAVs and human-driven vehicles (HDVs), on urban networks from various perspectives: traffic efficiency, environmental performance, and traffic safety. To this end, findings obtained from simulations of two real urban networks are employed to evaluate the impacts of penetration rates and different driving behaviors of CAVs. Six penetration rates (15%, 30%, 45%, 60%, 75%, and 90%), and three different driving behaviors of CAVs, namely defensive, normal, and aggressive, are considered in the scope of this study. While the Criteria Importance Through Intercriteria Correlation (CRITIC) method is utilized for determining objective criteria weights, the designed scenarios are scored by means of the Combined Compromise Solution (CoCoSo) method. The findings of the study indicate that defensive driving behavior enhances traffic safety, albeit with trade-offs in reduced traffic efficiency and increased traffic emissions. On the other hand, while aggressive driving behavior improves traffic efficiency and reduces traffic emissions, it also introduces safety risks, particularly at low penetration rates. According to the outcomes of the comprehensive evaluation, scenarios comprising only HDVs lose their dominance beyond 60% and 75% penetration rates, depending on the network. The proposed approach is expected to be effective in assessing the impacts of mixed traffic conditions on urban networks and can provide valuable insights to transportation policymakers and practitioners.

INDEX TERMS Multi-criteria analysis, mixed traffic, traffic microsimulation, traffic efficiency, traffic emissions, traffic safety.

I. INTRODUCTION

The advancements in communication and information technologies have brought substantial changes to the transportation sector, shaping both industry practices and research directions. This technological evolution has driven the accelerated adoption of intelligent transportation systems (ITS), underpinned by ongoing advancements and innovations [1], [2], [3]. One of the most widely examined advancements is the emergence of connected and autonomous vehicles (CAVs), which have captured significant interest from both academic scholars and industry professionals. CAVs, equipped with wireless communication technologies, possess the ability to communicate with surrounding entities such as other vehicles, traffic infrastructure and road users, enabling them to collect, process, and share data about their driving environments in real-time while autonomously navigating [4], [5], [6]. The deployment of CAV technology is anticipated to offer significant advantages across multiple aspects of transportation, including enhanced traffic safety, environmental benefits, and improved mobility [7]. However, implementing large-scale tests to verify the expected effects of CAVs presents challenges, primarily due to financial limitations and regulatory restrictions. Consequently, researchers are increasingly relying on traffic simulation models, especially microscopic traffic models, as a methodological tool to investigate and forecast the impacts of CAVs on traffic efficiency, environmental performance, and traffic safety

The associate editor coordinating the review of this manuscript and approving it for publication was Zhe Xiao.

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within the existing body of literature [8], [9], [10], [11], [12], [13], [14].

While CAVs hold the promise of significant improvements, it is estimated that by 2045, only up to 24.8% of the U.S. vehicle fleet will consist of CAVs [15]. Consequently, mixed traffic conditions, combining both CAVs and human-driven vehicles (HDVs) with their varied driving behaviors, will persist for a long period of time. It is assumed that the driving behaviors of CAVs will resemble those of HDVs. The distinctiveness of the driving behavior of CAVs needs to be identified, aligned with established human driving behaviors, such as defensive, normal, and aggressive [11]. Advancements in technology, especially in autonomy and connectivity, hold considerable potential to reshape vehicle behaviors across multiple dimensions. These impacts include changes in acceleration and deceleration patterns, modifications in vehicle-following dynamics, adjustments in lateral and gap acceptance behaviors, and overall improvements in decision-making capabilities. Unlike the inherently stochastic nature of human-driven vehicle behaviors, CAV behaviors can be systematically programmed and adjusted to reflect specific driving behaviors, such as defensive, normal, and aggressive [5], [16], [17], [18]. In this regard, this study incorporates three distinct driving behaviors of CAVs to assess their effects on two urban networks, using various evaluation measures related to traffic efficiency, environmental performance, and traffic safety.

In numerous fields, including transportation engineering, developing specific evaluation criteria is a standard practice for comparing and assessing alternatives. In this context, multi-criteria analysis (MCA) methods play a pivotal role by providing systematic frameworks for evaluating alternatives based on predefined criteria [19]. MCA methods offer rigorous and effective approaches for assessing the influence of various criteria, determining their respective weights, and conducting comprehensive scoring. Despite the advancements in CAV technology, the comprehensive evaluation of the impacts of mixed traffic conditions involving both CAVs and HDVs on urban networks remains underexplored. To the best of the authors' knowledge, no prior study has combined traffic microsimulation modeling with MCA methods to evaluate the impacts of different driving behaviors of CAVs on urban networks. Addressing this gap is crucial, as such an integration provides a novel and systematic approach for tackling the complex challenges associated with mixed traffic conditions. The novelty of this study lies in its pioneering integration of MCA methods with traffic microsimulation modeling to comprehensively assess how different driving behaviors of CAVs influence urban traffic networks. The key contributions and novelties of this study are three-fold: (1) this study is the first to combine MCA methods with traffic microsimulation modeling to analyze the impacts of different driving behaviors of CAVs on urban networks, with an emphasis on traffic efficiency, environmental performance, and traffic safety aspects; (2) the integration of MCA methods with traffic microsimulation modeling provides a systematic and multidimensional approach, enabling a comprehensive evaluation of various objectives, rather than focusing on the evaluation of one aspect as addressed in most existing literature; (3) the proposed methodology has the potential to provide valuable insights to transportation policymakers and practitioners for the prioritization of driving behaviors of CAV deployment on urban networks. An open-source simulation tool called Simulation of Urban Mobility (SUMO) is utilized as a simulation environment [20]. The Criteria Importance Through Intercriteria Correlation (CRITIC) method is used to determine objective criteria weights, and various designed scenarios are scored by means of the Combined Compromise Solution (CoCoSo) method [21], [22].

Following the introduction, Section II presents a recent review of the literature pertaining to the research topic. The research methodology, which includes the modeling behavior of CAVs, selected evaluation measures, and the applied MCA methodology, is outlined in Section III. Section IV introduces the case studies, including the calibration of simulation models and a description of the adopted scenario designs. The outcomes of the simulation models and multi-criteria analysis, as well as the sensitivity analysis, are presented in Section V. The final section provides conclusions and future research directions.

II. LITERATURE REVIEW

The literature review in this section is divided into four parts. The first part presents a concise review of studies pertaining to the influence of CAVs through simulation-based analyses. The second part examines studies that combine the use of MCA methods and traffic simulation studies. The third part of this section presents modeling of different behaviors of CAVs based on recent studies and field projects. The last part presents a summary of the literature and highlights the contributions of the paper.

A. SIMULATION STUDIES RELATED TO IMPACTS OF CAV

In the past few years, there has been an increased emphasis on modeling the behavior of CAVs within a simulation environment in the literature. The potential effects of CAV technology on traffic can be broadly categorized into three main areas: traffic efficiency, environmental impact, and traffic safety.

The study [14] discussed the effects of CAVs on both traffic efficiency and safety, taking into account different types of road networks and penetration rates of CAVs. The study validates that significant enhancements in traffic efficiency occur at lower levels of CAV implementation (approximately between 20% and 40%), but the specific range differs based on network type and traffic conditions. Furthermore, the safety findings indicate that the number of conflicts initially rises at lower penetration rates, but decreases considerably at higher rates. Another study [23], presented a theoretical method aimed at evaluating the impact of implementing a dedicated CAV lane (DL) policy on freeway traffic efficiency. The findings reveal that the optimal implementation of a CAV

DL policy is contingent upon several factors, including the market penetration rate of CAVs, platoon intensity, and the DL access control policy. The research outcomes provide valuable insights into the efficiency of traffic flow when considering the interaction between CAVs and HDVs under varying circumstances, thereby offering informed recommendations for the prudent utilization of CAV DL policies to enhance traffic efficiency. Another research [18] examined the impact of various driving behaviors of CAVs on the road capacity using fundamental diagram. The findings of this study indicate that the behavior of CAVs plays a significant role in affecting road capacity. Notably, adopting aggressive driving behavior by CAVs has the potential to enhance the capacity of standard freeways by up to 101%. However, the study also highlighted that if CAVs exhibit defensive behavior, it can lead to a reduction in freeway capacity by as much as 34%.

Within the context of research focused on traffic emissions and fuel consumption impacts, the study [24] conducted a comprehensive assessment of the influence of CAVs on fuel consumption and emissions within mixed traffic flow on expressways. According to the findings of this study, CAVs can greatly reduce fuel consumption and traffic emissions, resulting in reductions of up to 24.33% for HC, 27.06% for NO_x, 37.53% for CO, and 40.58% for fuel consumption with full adoption. Another study [25] investigated the impact of automated vehicles with varying degrees of automation on pollutant emissions. SUMO was used as the microscopic simulation environment. Automated vehicles can reduce CO emissions by 38.56%, CO_2 emissions by 17.09%, HC emissions by 36.3%, PM_x emissions by 28.12%, and NO_x emissions by 19.78%, respectively in the most optimistic scenario at the network level. The studies [26], [27] analyzed the network-wide emission effects of vehicles with Cooperative Adaptive Cruise Control (CACC) using hypothetical road networks.

Within the context of research focused on traffic safety impacts, the study [28] investigated the influence of mixed traffic conditions, including CAVs and HDVs, on traffic safety. The findings demonstrate that the use of a more cautious car following strategy of CAVs would result in a significant improvement in traffic safety, albeit with a comparatively smaller enhancement in traffic capacity. As the CAV penetration rate increases, stop-and-go traffic is significantly alleviated. Another study [29] developed a decision-making CAV control algorithm in VISSIM to evaluate safety impacts on a motorway corridor. This study highlighted the potential safety benefits of CAVs in improving road safety, especially at high market penetration rates. The observed reduction varied with different levels of CAV penetration, with percentages ranging from 12% to 47% for a 25% penetration rate, 50% to 80% for a 50% penetration rate, 82% to 92% for a 75% penetration rate, and 90% to 94% for a 100% penetration rate.

In sum, the studies reviewed in this part emphasize the significant potential of CAVs to improve traffic efficiency, reduce traffic emissions and fuel consumption, and enhance traffic safety, while also illustrating that these benefits rely on various factors such as penetration rates, network types, and driving behaviors. The synthesis of findings from the reviewed studies highlights the importance of adopting a comprehensive evaluation approach that incorporates multiple aspects to effectively realize the potential benefits of CAV deployment.

B. COMBINED USE OF MCA METHODS AND TRAFFIC SIMULATION MODELS

Criteria weighting methods in MCA applications can be categorized into three distinct groups: subjective weighting, objective weighting, and combined weighting methods. Subjective weighting methods entail the incorporation of expert judgment and evaluation throughout the process. In contrast, objective weighting methods depend on the mathematical assessment of criteria values. Unlike subjective methods, objective weighting methods do not necessitate decision makers to explicitly articulate their preferences regarding the evaluation criteria. As objective methods derive criteria weights solely from mathematical functions, any subjectivity in the decision-making process, which may arise from differing opinions among decision-makers, is completely removed. The combined approach is a combination of subjective and objective methods [30]. Although various studies have employed subjective weighting methods in the transportation engineering discipline [31], [32], [33], [34], it is important to emphasize that all the reviewed studies in this section employ a combination of objective weighting methods and traffic microsimulation modeling.

The study [35] examined sustainable design schemes for three-lane entrance ramps on expressways. This study presented a comprehensive evaluation (CE) method that integrates the CRITIC method and the Technique for Order Preference by Similarity to an Ideal Solution (TOPSIS) method to recommend optimal designs for various traffic scenarios. Another study [36] proposed a comprehensive management strategy that encompasses all road users, with particular attention to cyclists due to their increasing presence on two-lane rural roads. This study utilized a traffic microsimulation model to simulate various speed limits on a narrow rural road. The simulations considered different scenarios that combined motorized and bicycle traffic levels. Evaluation measures were then derived from these simulations to assess the safety and traffic flow characteristics. The optimal speed limit for each traffic level was determined by employing the CRITIC method in conjunction with the simulation outputs.

The study [37] evaluated the operational features and the selection of configurations for multilane turbo roundabouts by utilizing the entropy weighting method. The entropy weight method is employed to determine the weights of selected indicators, enabling a comprehensive evaluation of various roundabout designs across different traffic scenarios. Additionally, the operational characteristics of these

roundabouts are modeled and analyzed using the VISSIM software. Another study [38] presented an evaluation and analysis model for determining the optimal lengths of added direct left-turn (DLT) lanes using the entropy evaluation method. The model integrates simulations conducted with VISSIM and the Surrogate Safety Assessment Model (SSAM) software. A systematic design process for the determination of DLT lane lengths is proposed based on the findings from the model in the scope of this study. Another study [39] introduced a multi-objective evaluation approach to assess the performance of different midblock crosswalk treatments on urban roadways. The TOPSIS entropy evaluation method is employed to rank the performance of various crosswalk treatments, taking into account multiple performance metrics.

In brief, the reviewed studies reveal a clear trend of integrating simulation tools with MCA methods to evaluate diverse aspects of transportation engineering. The consistent application of these methodologies demonstrates their effectiveness and reliability in providing valuable insights for transportation policymakers and practitioners.

C. BEHAVIOR MODELING OF CAVs

In microscopic traffic simulation models, different driving behaviors are represented through longitudinal and lateral configurations. Longitudinal dynamics are modeled using car-following models, while lateral dynamics are captured through lane-changing models [5]. Technological advancements in autonomy and connectivity may influence vehicle behavior, potentially leading to changes in acceleration and deceleration patterns, as well as modifications in car-following and lane-changing behaviors. While human-driven vehicle behaviors exhibit inherent stochastic characteristics, the behaviors of CAVs are programmable, enabling them to be tailored to specific driving behaviors, such as defensive, normal, and aggressive [16], [40]. Defensive driving behavior entails adherence to road regulations and a focus on traffic safety. A vehicle employing defensive driving maintains larger headway values and moves more cautiously than a typical human driver. Normal driving behavior resembles that of human-driven vehicles and incorporates onboard sensors to measure the distances and speeds of nearby vehicles. In contrast, aggressive driving behavior maintains smaller headway values while assuming perfect perception and prediction. Furthermore, a vehicle employing aggressive driving exhibits more aggressive speed changes while engaging in collaborative actions with nearby vehicles [11], [16], [17], [18]. Various studies have investigated the different driving behaviors exhibited by CAVs [11], [16], [17], [18]. In the CoEXist project [17], three distinct driving behaviors, namely cautious, normal, and all-knowing, were proposed. To calibrate the parameters associated with these behaviors, real-world data were obtained from three Toyota Prius vehicles equipped with customized driving behaviors. Furthermore, a comprehensive classification scheme was proposed to categorize and analyze different driving behaviors exhibited by CAVs in the study [16]. This scheme consists of nine discrete levels, representing a broad spectrum of driving behaviors, with Level 1 denoting the most cautious approach and Level 9 indicating the most aggressive driving behavior. The study [41] adopted the recommended nine-level driving parameters and proposed a methodology aimed at optimizing the performance of mixed traffic conditions within a freeway network. Another study [42] utilized the calibrated parameters of the CoEXist project to reveal the influence of CAVs on traffic operations at roundabouts. In summary, the reviewed studies underscore the significance of modeling diverse driving behaviors of CAVs within traffic simulations to accurately predict and assess their effects on real-world traffic scenarios. As CAV technology progresses, the capability to simulate and adapt these behaviors to varying traffic conditions will play a pivotal role in facilitating their safe and efficient integration into existing transportation systems.

D. SUMMARY OF LITERATURE AND CONTRIBUTION OF THE PAPER

The preceding sections have provided a review of the current state of research in the areas of CAV impact simulation studies, the integration of MCA methods with traffic simulation models, and the behavior modeling of CAVs. Table 1 provides a summary of state-of-the-art papers encompassing all sections of the literature review.

TABLE 1. The recent studies encompassing all sections of the literature review.

Study	Study Area/Model type	Evaluation
[43]	Eastern Sydney urban network/Microsimulation model using VISSIM tool	Travel time reliability
[44]	Expressway in Beijing/ Microsimulation model using VISSIM tool	Traffic efficiency and energy consumption
[45]	Three cloverleaf interchanges in Louisiana/ Microsimulation model using VISSIM tool	Traffic operation and safety
[46]	Hypothetical grid network/ Microsimulation model using SUMO tool	Traffic operation and safety
[47]	Expressway network in Shanghai/ Multiclass cell transmission model	Traffic emissions and fuel consumption
[48]	Multiple traffic scenarios/ Microsimulation model using SUMO tool	Traffic emissions and fuel consumption
[49]	M50 motorway in Dublin/Macroscopic model	Average velocity and level of service
[50]	Expressway in China/ CRITIC method combined with microsimulation model	Traffic efficiency, environmental impact and safety
[51]	Compound interchange in China/Comprehensive analysis with microsimulation model	Traffic efficiency, environmental impact and safety
[52]	Hypothetical grid network/ Microsimulation model considering different driving behaviors of CAVs	Traffic efficiency, environmental impact and safety

Previous studies have explored the impacts of CAVs on traffic efficiency [14], environmental performance [24], [25], and traffic safety [28], [29] for diverse traffic environments. However, the combination of these aspects using a comprehensive evaluation framework remains underexplored. In this regard, this study integrates traffic microsimulation modeling with MCA techniques to thoroughly examine the impacts of different driving behaviors of CAVs on the urban network in terms of traffic efficiency, environmental performance, and traffic safety. The combined use of MCA methods and traffic microsimulation modeling presents a comprehensive approach that allows for the evaluation of multiple aspects, as opposed to the limitations of single criterion evaluations. This study not only contributes to the understanding of the impacts of CAVs but also provides valuable insights to transportation policymakers and practitioners for the prioritization of driving behaviors of CAV deployment on urban networks. The proposed methodology has the potential to inform the design and management of future traffic systems in a manner that optimizes the benefits of CAVs while mitigating potential challenges.

III. METHODOLOGY

This study introduces a comprehensive evaluation framework that integrates multi-criteria analysis approaches with traffic microsimulation modeling for assessing mixed traffic conditions on urban networks. The novelty of this approach lies in its structured combination of simulation-based traffic analysis and advanced MCA methods, allowing for a comprehensive assessment of mixed traffic conditions in terms of traffic efficiency, environmental performance, and traffic safety aspects. To ensure objectivity in criteria weighting, the CRITIC method is applied, while the CoCoSo method is employed for obtaining the comprehensive scores of scenarios. The synergy of these methods enhances the objectivity, robustness, and reliability of the evaluation process. The application of the CRITIC weighting method in traffic simulation studies remains highly limited in the literature, and this study also contributes to this research field. Furthermore, the impact of changing objective weighting method and applying different MCA methods on the study's findings is examined through sensitivity analysis. First, the Entropy method is employed to compare the criteria weights obtained from the CRITIC method and to assess the scenarios using the CoCoSo method. Secondly, the findings derived from the CRITIC-CoCoSo method are compared with those obtained through alternative MCA approaches, namely the Preference Ranking Organization Method for Enrichment of Evaluations II (PROMETHEE II) and the Multi-Attributive Border Approximation Area Comparison (MABAC) method. The two-stage sensitivity analysis conducted in this study provides a systematic approach to evaluating the robustness of the findings.

The methodology is structured into three main stages: modeling the different behaviors of CAVs, selection of evaluation measures and scenario configurations, and analysis of simulation results utilizing MCA methods. The flowchart of the methodology is illustrated in Figure 1. Figure 1 depicts the steps of the methodology, starting with data collection from urban networks. As the second step, the calibration of baseline simulation models is conducted, and the evaluation measures are selected, encompassing traffic efficiency, environmental performance, and traffic safety aspects based on a literature review. The simulation of adopted scenarios is conducted using the SUMO simulation tool. The simulation results are analyzed with MCA methods to evaluate the performance of the scenarios comprehensively. The sensitivity analysis is performed in the last step.

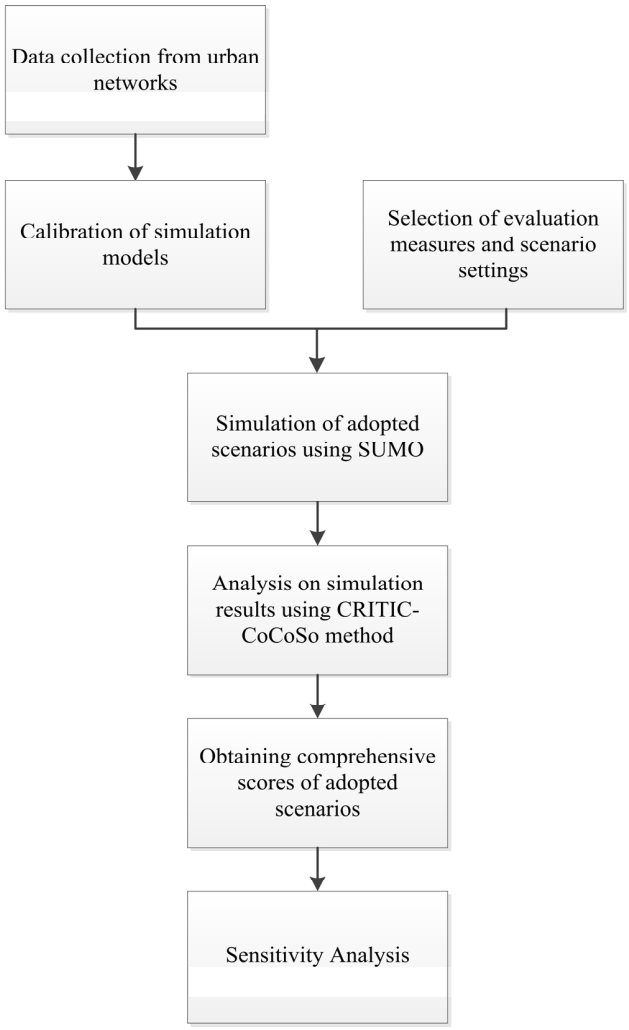


FIGURE 1. The flowchart of the methodology.

A. MODELING DIFFERENT DRIVING BEHAVIORS OF CAVs

In the scope of this study, four different vehicle types are represented: human-driven vehicles (HDVs), CAV-defensive, CAV-normal, and CAV-aggressive. Microscopic traffic models govern the behavior of vehicles through car-following, lane-changing, and gap-acceptance models. A realistic and comprehensive representation of CAVs' behavior is needed

by incorporating methods from robotic simulation, which can lead to an excessively time-consuming computational load. This approach is not suitable for large-scale microscopic traffic simulations that involve hundreds of vehicles at the same time [53]. In this context, the driving abstraction technique is employed within the scope of this study. The driving abstraction technique refers to replicating the changes in vehicle movements that occur as a result of implementing technology, rather than considering the actual deployment of the technology [41]. The automation level of CAVs in this study is considered to be fully automated vehicles, which possess the capability of performing driving tasks in diverse environmental and road conditions [18]. The parameters of the simulation model, which include car-following, and lane changing model parameters, are adjusted to model the different behaviors of CAVs.

To model different behaviors of CAVs discussed above, an open-source simulation tool called SUMO is utilized. The default car-following model in SUMO is the Krauss car-following model, which is utilized for modeling the longitudinal movement of HDVs. The details of the Krauss car-following model can be found in the study [54]. For modeling the longitudinal behavior of CAVs, the Intelligent Driver Model (IDM) is employed [55]. It is widely employed for modeling the behavior of CAVs in the literature [56], [57]. The default lane-changing model in SUMO is the LC2013 model, which employs the categorization of the causal factors contributing to lane-changing behavior. These factors are classified into strategic, cooperative, tactical, and obligatory reasons. The LC2013 lane-changing model is used for both HDVs and CAVs [58].

The parameters of the car-following and lane-changing model are changed to reflect the different driving behaviors of CAVs and HDVs. This approach is widely used in the literature for modeling the differences in the behavior of CAVs and HDVs [5], [10], [16], [17], [42]. The parameters for the car-following and lane-changing models utilized in this study are sourced from literature studies that present different driving behaviors of CAVs ranging from defensive to aggressive. It is reasonable that defensive driving behavior is characterized by greater headway, minimum gap, and lower acceleration, whereas normal and aggressive driving behavior tend to exhibit smaller headway, minimum gap, and higher acceleration [18]. In this regard, the parameters, such as ''minGap'', ''tau'', ''accel'' and ''decel'' of the car-following model are adjusted based on the vehicle type. The ''tau'' parameter in SUMO denotes the driver's desired (minimum) time-headway. It is set to 2.1 s for CAV-defensive, 0.9 s for CAV-normal and 0.6 s for CAV-aggressive vehicles. The ''minGap'' parameter in SUMO denotes the minimum gap when standing in meters. It is set to 2.5 m for CAV-defensive, 1.5 m for CAV-normal and 1.0 m for CAV-aggressive vehicles. While the parameter ''accel'' denotes the acceleration capability of vehicles, the parameter ''decel'' denotes to their deceleration capability. The value of the ''accel'' parameter is set to 3.1 for CAV-defensive, 3.5 for CAV-normal and 3.9 for CAV-aggressive vehicles. The ''decel'' and ''emergencyDecel'' values are set the same for all vehicle types to avoid collision. The ''lcAssertive'' parameter of the LC2013 lane-changing model in SUMO denotes the eagerness to accept lower front and rear gaps on the target lane, and it is adjusted according to vehicle type. When the ''lcAssertive'' parameter value is higher, vehicles accept lower gaps for lane-changing. It is set to 0.6 for CAV-defensive, 0.7 for CAV-normal and 0.8 for CAV-aggressive vehicles [14], [16], [18], [59]. The explanations of the used model parameters for different driving behaviors of CAVs are presented in Table 2. The remaining model parameters are employed with default values in SUMO. The used model parameters of HDVs are provided in the calibration of the simulation model section.

TABLE 2. The detailed model parameters used for the different driving behaviors of CAVs.

Parameters	CAV-defensive	CAV-normal	CAV-aggressive
tau (s)	2.1	0.9	0.6
minGap (m)	2.5	1.5	1.0
accel (m/s^2)	3.1	3.5	3.9
decel (m/s^2)	4.5	4.5	4.5
emergencyDecel (m/s^2)	8.0	8.0	8.0
lcAssertive	0.6	0.7	0.8

tau: The driver's desired (minimum) time-headway
minGap: The minimum gap when standing
accel: The acceleration ability of the vehicle
decel: The deceleration ability of the vehicle
emergencyDecel: The maximum deceleration ability of the vehicle in case of emergency
lcAssertive: The eagerness to accept lower front and rear gaps on the target lane

B. EVALUATION MEASURES

To conduct a comprehensive evaluation, the chosen measures for evaluating the outcomes of the simulation need to encompass various aspects, including traffic efficiency, environmental performance, and traffic safety. The evaluation measures are taken into account based on the literature within the scope of this study [31], [32]. The selected evaluation measures are labeled as C1, C2, C3, C4, and C5.

C1-Average speed (km/hr): In terms of traffic efficiency, the average speed of all vehicles is considered.

C2-Average time-loss (s): In terms of traffic efficiency, the average time-loss per vehicle is considered. Time-loss is calculated for each vehicle in the simulation and refers to the difference between the actual trip duration and the theoretical trip duration in the SUMO simulation tool.

C3-Average CO emission (g): In terms of environmental performance, the average CO emission measure is considered.

C4-Average fuel consumption (g): Another evaluation measure, in terms of environmental performance, average fuel consumption, is taken into account.

C5-Total number of conflicts: To evaluate the impact of traffic safety, the total number of conflicts is taken into account using a combination of time-to-collision (TTC) and post-encroachment time (PET) thresholds.

C. MULTI-CRITERIA ANALYSIS (MCA)

Multi-criteria analysis (MCA) methods represent a systematic and structured approach that evaluates the various objectives in a comprehensive manner. Instead of focusing on the evaluation of one aspect, these methods help to consider a range of objectives simultaneously [34], [60]. In the context of evaluating the impact of different driving behaviors of CAVs on urban networks, using a multi-criteria analysis approach allows for a structured and systematic assessment of how various behaviors affect traffic efficiency, environmental performance, and traffic safety aspects. To evaluate the impact of different driving behaviors of CAVs on urban networks, the proposed methodology incorporates three primary objectives: i) enhancing traffic efficiency, ii) improving traffic safety, and iii) reducing the harmful effects of environmental outcomes. To address these objectives within the framework, a multi-criteria analysis approach is utilized. While the weights of the selected evaluation measures are calculated using an objective weighting method called CRITIC, the evaluation of the designed scenarios is conducted using the CoCoSo method in the context of this study. Road transportation is widely acknowledged as a complex system, comprising interdependent factors across multiple dimensions. The CRITIC method is particularly suited for this context, as it effectively accounts for the variability and correlation present in traffic data. This method is chosen for its capability to objectively determine criteria weights based on data variability and correlation, making it suitable for our multi-criteria analysis [21], [36]. The CoCoSo method is chosen for its ability to efficiently account for conflicting evaluation criteria, and its effectiveness in combining diverse perspectives to provide a robust evaluation of the scenarios [22].

Contrary to other objective weighting methods such as entropy and standard deviation methods, the CRITIC method utilizes volatility and conflict within the data by incorporating two indicators:contrast intensity and correlation. The contrast intensity of each decision criterion is measured using standard deviation (SD) across different alternatives. A higher SD indicates that this criterion has greater variation between the alternatives, and thus it is assumed that it presents meaningful information, leading to a higher weight being assigned to this criterion. On the other hand, the CRITIC method accounts for the conflict relationship between criteria through the use of a linear correlation coefficient, where values close to -1 signify strong conflict between criteria and values near 1 indicate a parallel relationship, implying redundancy. A higher correlation coefficient indicates a lower level of conflict, and, as a result, a lower weight is assigned [21], [35], [36]. The application steps of the CRITIC method are as follows:

Step-1: The decision matrix, called X, is formed to indicate the performance of different alternatives using simulation results in relation to various criteria.

$$X = [x_{ij}]_{mxn} = \begin{bmatrix} x_{11} & \cdots & x_{1n} \\ \vdots & \ddots & \vdots \\ x_{m1} & \cdots & x_{mn} \end{bmatrix}; \quad (i = 1, 2, \ldots., m \text{ and } j = 1, 2, \ldots n) \tag{1}$$

x_{ij} present the performance value of ith alternative on the jth criterion, m is the alternative, n is the evaluation criteria.

Step-2: Normalization of decision matrix

$$x_{ij}^{*} = \frac{x_{ij} - x_j^{worst}}{x_j^{best} - x_j^{worst}} \tag{2}$$

x_j^{best} and x_j^{worst} represent the best and worst scores of criterion j. The identification of best and worst scores for each criterion hinges on its inherent characteristics, specifically whether it is beneficial or not.

Step-3: The determination of criteria weights involves the consideration of two factors: the standard deviation of the criterion and its correlation with other criteria. To assess the criteria contrast, the standard deviation of the criterion values is first normalized by column. In addition, the correlation coefficients between all pairs of columns are taken into account. The weight of the jth criterion (w_j) is calculated as:

$$C_j = \sigma_j \sum_{j=1}^{n} (1 - r_{jj'}) \tag{3}$$

$$w_j = \frac{C_j}{\sum_{j=1}^{n} C_j} \tag{4}$$

While C_j denotes the quantity of information contained in jth criterion, σ_j denotes the standard deviation of the jth criterion and $r_{jj'}$ denotes the correlation coefficient between the jth andj'th criteria. A criterion that has a high standard deviation and low correlation with other criteria indicates that the weight assigned to the given criterion is high [61].

The CoCoSo method was proposed in the study [22]. This method integrates the weighted sum approach and the exponentially weighted product approach, offering a comprehensive framework for generating compromise solutions. Thanks to its use of different aggregator strategies to form a robust evaluation, this method is widely adopted for addressing various decision-making problems in the transportation engineering field [62], [63]. The weight values calculated from the CRITIC method are utilized in the CoCoSo method. The application steps of the CoCoSo method are outlined as follows:

Step-1: The decision matrix is formed as in Eq. (5).

$$X = [x_{ij}]_{mxn} = \begin{bmatrix} x_{11} & \cdots & x_{1n} \\ \vdots & \ddots & \vdots \\ x_{m1} & \cdots & x_{mn} \end{bmatrix}; \quad (5)$$

x_{ij} represent the performance value of ith alternative on jth criterion, m is the alternative, n is the evaluation criteria.

Step-2: Normalization of decision matrix

For the normalization process, a classification is carried out according to the beneficial or non-beneficial criteria.

$$r_{ij} = \frac{x_{ij} - min_i x_{ij}}{max_i x_{ij} - min_i x_{ij}}; for\ benefit\ criteria \quad (6)$$

$$r_{ij} = \frac{max_i x_{ij} - x_{ij}}{max_i x_{ij} - min_i x_{ij}}; for\ cost\ criteria \quad (7)$$

Step-3: Determine the total of weighted comparability weighted sequence (S_i) and power weighted (P_i) comparability sequences as follows:

$$S_i = \sum_{j=1}^{n} w_j r_{ij} \quad (8)$$

$$P_i = \sum_{j=1}^{n} (r_{ij})^{wj} \quad (9)$$

Step-4: Calculating the relative weights for alternatives using three distinct appraisal score methods:

$$k_{ia} = \frac{P_i + S_i}{\sum_{i=1}^{m} (P_i + S_i)} \quad (10)$$

$$k_{ib} = \frac{S_i}{min_i S_i} + \frac{P_i}{min_i P_i} \quad (11)$$

$$k_{ic} = \frac{\lambda S_i + (1-\lambda) P_i}{\lambda max_i S_i + (1-\lambda) max_i P_i}; 0 \le \lambda \le 1 \quad (12)$$

λ value is generally determined to be 0.5 [22].

Step-5: Calculating the final score for each alternative using:

$$k_i = (k_{ia} k_{ib} k_{ic})^{\frac{1}{3}} + \frac{1}{3}(k_{ia} + k_{ib} + k_{ic}) \quad (13)$$

The algorithm of the CRITIC-CoCoSo method is presented in Table 3.

TABLE 3. The algorithm of the CRITIC-CoCoSo method.

Sequential algorithm of the CRITIC-CoCoSo method
Input: Decision matrix denoted as *(X)*, with set of alternatives (designed scenarios) and criteria (evaluation measures)
Output: Final score of scenarios (k_i)
Begin
1. Normalize decision matrix X in CRITIC method // see Eq.(2)
2. Compute weights (w_j) for each criterion j // see Eqs.(3-4)
3. Normalize decision matrix X in CoCoSo method // see Eqs.(5-7)
4. Compute the total of weighted comparability weighted sequence (S_i) and power weighted (P_i) comparability sequences for each scenario i // see Eqs.(8-9)
5. Compute final score of scenarios (k_i) // see Eqs.(10-13)
End

IV. CASE STUDY

In the context of this study, the proposed methodology is tested through two real-world urban networks with field measurements. The first urban network is located in Balıkesir (hereafter called Network-A), and the second is located in the Gebze district (hereafter called Network-B) in Türkiye.

Network-A consists of two consecutively located signalized intersections and encompasses a road network approximately 5 km in length. Network-B includes signalized intersections and a roundabout, and has approximately 4-km road network length. Both networks are close to the city center and are located in densely populated residential areas. Network-A and Network-B are selected to represent realistic urban traffic scenarios with varying levels of size and complexity. Network-A represents a simple urban arterial corridor with traffic signal control. On the other hand, Network-B integrates signalized intersections and a roundabout, accommodating diverse movement patterns such as merging and diverging, thereby introducing greater complexity and interaction dynamics. Traffic simulation models help analyze and predict transportation system performance, aiding in the planning, design, and optimization of traffic operations. These models are classified into macroscopic, mesoscopic, and microscopic categories based on their level of detail. Macroscopic models describe traffic flow characteristics such as density, flow, and speed, while mesoscopic models provide a moderate level of detail between macroscopic and microscopic models. Microscopic models deal with the simulation of the movement of individual vehicles utilizing car-following and lane-changing theories [64]. In this study, microsimulation modeling is employed using the SUMO simulation tool, as it focuses on the detailed interactions of individual vehicles to accurately capture traffic flow dynamics.

The aerial image and created model in SUMO of Network-A and Network-B are illustrated in Figure 2 and Figure 3, respectively. Figure 2 and Figure 3 provide aerial views of Network-A and Network-B, respectively, highlighting their structural layouts. The corresponding SUMO model is also depicted, demonstrating how the network's geometries were accurately represented in the simulation environment. Both signalized intersections in Network-A have an identical cycle length, which is 134 s. The signal control program consists of four phases, and each approach has a separate green duration. At Intersection-I, the green signal duration is 37 s for the northbound approach, 18 s for the westbound approach, 19 s for the eastbound approach, and 28 s for the southbound approach, respectively. At Intersection-II, the green signal duration is 32 s for the northbound approach, 19 s for the westbound approach, 15 s for the eastbound approach, and 36 s for the southbound approach, respectively. The cycle length of signalized intersection in Network-B is 90 s, each approach has 37 s green light duration. The duration of yellow and all-red interval are 4 s for both network. Figure 4 presents the phase diagrams of signalized intersections in Network-A

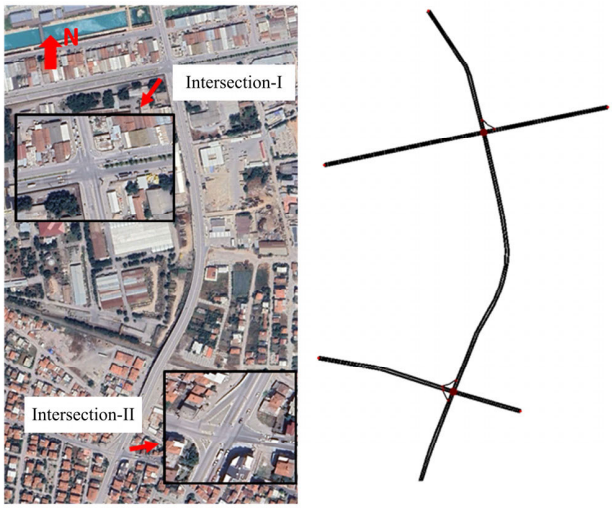


FIGURE 2. The aerial image and created model in SUMO of Network-A.

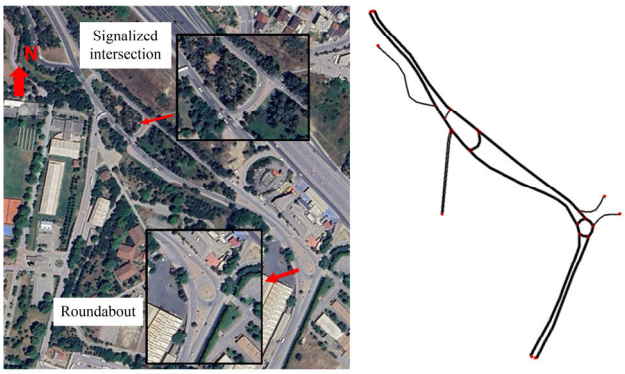


FIGURE 3. The aerial image and created model in SUMO of Network-B.

and Network-B. The direction of vehicular movement is indicated by black arrows in the figure.

A. CALIBRATION OF SIMULATION MODELS

To establish a reliable simulation model, it is imperative to carry out the calibration of the simulation model. The primary objective of calibration is to reduce the discrepancy between the observed measurements and the simulation model [65]. For this purpose, the measures selected in the scope of this study are the GEH Statistic and Theil's U-coefficient with its bias (U_m), variance (U_s) and covariance (U_c) elements [66]. The bias element (U_m) signifies the presence of systematic error between the observed and simulated data. The variance element (U_s) elucidates the variability in the observed and simulated data. The covariance element (U_c) denotes the residual error beyond the influences of bias and variance [67]. GEH statistic is formulated in Eq. (14), where E represents the simulated traffic flow, and V represents the observed

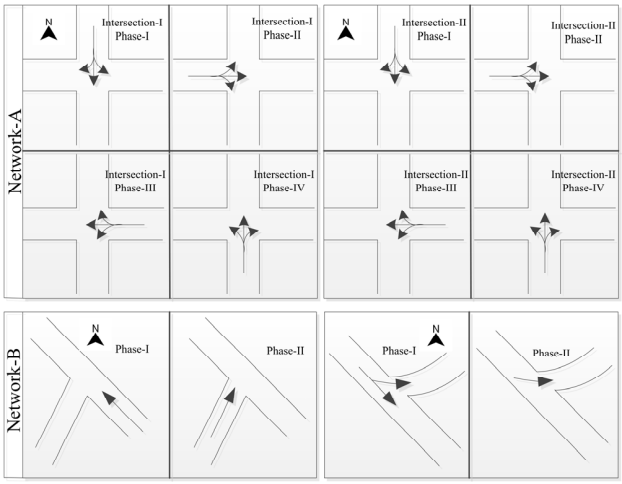


FIGURE 4. The phase diagrams of signalized intersections in Network-A and Network-B.

traffic flow. The other calibration measure considered is Theil's U-coefficient with its components, defined between Eqs. (15)-(18), where x_i denotes the simulated measurement, y_i denotes the observed measurement, $\bar{x}_i$ denotes the mean of simulated measurement, $\bar{y}_i$ denotes the mean of observed measurement, σ_x denotes the standard deviation of simulated measurement, σ_y denotes the standard deviation of the observed measurement, r denotes the coefficient of correlation and N denotes the number of measurements. While $U = 0$ represents the perfect fit, $U = 1$ represents the worst fit. On the other hand, it is deemed favorable when the values of U_m and U_s are closer to 0. It is evaluated as good, if the value of U_c is closer to 1 [67].

$$GEH = \sqrt{\frac{2(E-V)^2}{E+V}} \tag{14}$$

$$U = \frac{\sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - x_i)^2}}{\sqrt{\frac{1}{N}\sum_{i=1}^{N}y_i^2} + \sqrt{\frac{1}{N}\sum_{i=1}^{N}x_i^2}} \tag{15}$$

$$U_m = \frac{(\bar{y}_i - \bar{x}_i)^2}{\frac{1}{N}\cdot\sum_{i=1}^{N}(y_i - x_i)^2} \tag{16}$$

$$U_s = \frac{(\sigma_x - \sigma_y)^2}{\frac{1}{N}\cdot\sum_{i=1}^{N}(y_i - x_i)^2} \tag{17}$$

$$U_C = \frac{2(1-r)\sigma_x\sigma_y}{\frac{1}{N}\cdot\sum_{i=1}^{N}(y_i - x_i)^2} \tag{18}$$

During the calibration process, the objective is to minimize the discrepancy between the measured and simulated traffic flows. For Network-A, the field data input is given to the simulation model for one-hour morning peak period, and for Network-B, the field data input is given to the simulation model for a two-hour evening peak period. The default car-following model, Krauss, and the lane-changing model, LC2013, are used to model the behavior of HDVs. The GEH statistic is calculated utilizing the observed traffic flow

through a field survey and the obtained traffic flow from the simulation results of Network-A and Network-B. The distribution of GEH values for links considered in Network-A and Network-B is plotted in Figure 5 and Figure 6, respectively. Furthermore, the GEH statistic for the sum of all link flows is calculated as 1.50 for the morning peak hour (8:00 A.M. – 9:00 A.M.) in Network-A. In Network-B, it is calculated as 1.25 for (5:00 P.M – 6:00 P.M), and 1.16 for (6:00 P.M. – 7:00 P.M.). As observed in Figure 5 and Figure 6, GEH statistics are calculated as lower than 5 for all links in both networks. According to the recommendation of the Federal Highway Administration (FHWA), a GEH value below 5 for over 85% of individual link flows and less than 4 for the sum of all link flows is deemed satisfactory [68].

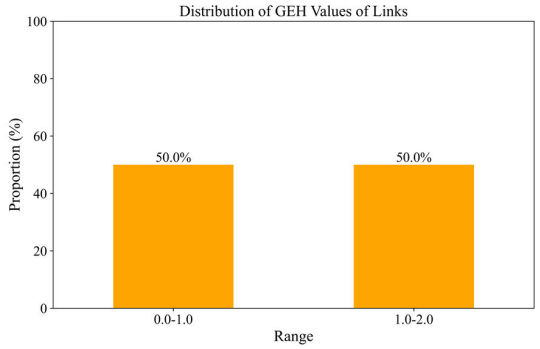


FIGURE 5. Distribution of GEH values of links in Network-A.

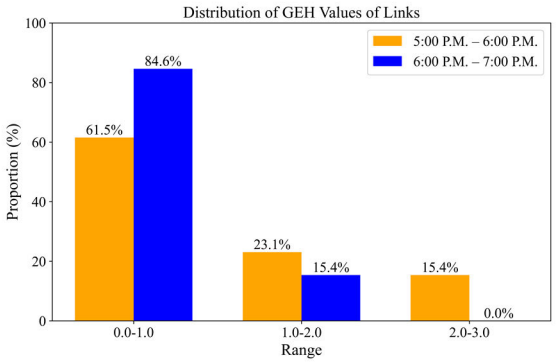


FIGURE 6. Distribution of GEH values of links in Network-B.

The Theil's U-coefficient, with its components for both networks, is given in Table 4.

TABLE 4. The calculated values of Theil's U-coefficient with its components for both network.

Theil's coefficients	U	U_m	U_s	U_c
Network-A	0.028	0.097	0.001	0.902
Network-B	0.016	0.086	0.009	0.905

The calculated values of the U-Theil's coefficient with its components given in Table 4, are acceptable [66], [67]. The analysis of the U-Theil coefficient for both networks demonstrates a strong model fit, with values of 0.028 for Network-A and 0.016 for Network-B, indicating minimal deviation from perfect fit ($U = 0$). The components U_m and U_s are favorable in both networks, with values close to zero, suggesting low systematic bias and variability between the observed and simulated data. Additionally, the high values of U_c, 0.902 for Network-A and 0.905 for Network-B, indicate good model fit. It is considered good if the covariance component (U_c) is near to 1 [66], [67]. Consequently, the findings indicate that the simulation model results closely match the field data. Having calibrated the simulation model in SUMO, the obtained model parameters of HDV are presented in Table 5.

TABLE 5. The model parameters of HDV.

Parameters	HDV
tau (s): The driver's desired (minimum) time-headway	1.4
minGap (m): The minimum gap when standing	2.5
accel (m/s^2): The acceleration ability of the vehicle	2.6
decel (m/s^2): The deceleration ability of the vehicle	4.5
emergencyDecel (m/s^2): The maximum deceleration ability of the vehicle in case of emergency	8.0
sigma: The driver imperfection parameter	0.6

B. DESIGNED SCENARIOS AND SIMULATION SETTINGS

In this study, six different penetration rates are used to evaluate the impacts of different driving behaviors of CAVs on both networks: 15%, 30%, 45%, 60%, 75%, and 90% based on previous studies [14], [32]. These penetration rates are labeled as P1, P2, P3, P4, P5, and P6, respectively. The full penetration of CAVs is considered an extreme case and is not included in the scope of this study. First, the current baseline scenario, which has 100% HDVs, is simulated for both networks and labeled as Scenario-1. In mixed traffic condition scenarios, one type of CAV behavior and equal distributions of different behaviors of CAVs at each penetration rate are studied. The scenarios adopted for both networks are presented in Table 6.

As an example to provide a better understanding, Scenario-A1 refers to the simulation of 100% HDVs for Network-A, and Scenario-B1 refers to the simulation of 100% HDVs for Network-B. Scenario-A2 refers to the simulation of an 85% penetration rate of HDVs and 15% CAV-defensive vehicles for Network-A, while Scenario-B2 refers to the simulation of an 85% penetration rate of HDVs and 15% CAV-defensive vehicles for Network-B.

The Eclipse SUMO is an open-source traffic simulation software developed by the German Aerospace Center (DLR) and employed as the simulation platform in this study. The graphical network editor of SUMO, referred to as Netedit, serves the purpose of developing road networks. It possesses the proficiency to construct networks from scratch and

TABLE 6. The simulated scenarios design.

Network Configurations	Scenario	HDV	CAV def	CAV normal	CAV agg
Network-A Network-B	S-1	100%	-	-	-
	S-2	85%	15%	-	-
	S-3	85%	-	15%	-
	S-4	85%	-	-	15%
	S-5	85%	5%	5%	5%
	S-6	70%	30%	-	-
	S-7	70%	-	30%	-
	S-8	70%	-	-	30%
	S-9	70%	10%	10%	10%
	S-10	55%	45%	-	-
	S-11	55%	-	45%	-
	S-12	55%	-	-	45%
	S-13	55%	15%	15%	15%
	S-14	40%	60%	-	-
	S-15	40%	-	60%	-
	S-16	40%	-	-	60%
	S-17	40%	20%	20%	20%
	S-18	25%	75%	-	-
	S-19	25%	-	75%	-
	S-20	25%	-	-	75%
	S-21	25%	25%	25%	25%
	S-22	10%	90%	-	-
	S-23	10%	-	90%	-
	S-24	10%	-	-	90%
	S-25	10%	30%	30%	30%

modify all characteristics of existing networks. Traffic demand data is specified by means of code editor software in a legible format with SUMO [20]. The results of the scenarios are obtained using SUMO output files. The current default emission model of SUMO, called the Handbook Emission Factors for Road Transport (HBEFA3) model, with the following Euro norm 4, is utilized to reveal the environmental performance of different driving behaviors of CAVs. In the simulation of examined networks, it is assumed that the engine and exhaust characteristics of all vehicles adhere to the categorization of passenger cars equipped with internal combustion engines, as outlined in the HBEFA3 database. To ensure a fair comparison, it is also assumed that CAVs conform to the HBEFA3 model with the following Euro norm 4. In accordance with the framework of the model, the power demand of vehicles is determined through the application of Eq. (19), where c_n denotes the linear model estimation coefficients, v denotes speed, and a denotes the acceleration [25]. Further details on the emission model can be found in the study [69].

$$P = c_0 + c_1 va + c_2 va^2 + c_3 v + c_4 v^2 + c_5 v^3 \tag{19}$$

To evaluate the safety impacts of the different driving behaviors of CAVs, surrogate safety measures (SSMs), namely time-to-collision (TTC) and post-encroachment time (PET) metrics, are utilized to obtain the number of conflicts. TTC is a frequently applied traffic safety metric that is well-suited for identifying car-following types of conflicts, specifically potential rear-end collisions. It is defined as the amount of time remaining before a rear-end collision occurs between two vehicles in the same lane, assuming they are maintaining their speed. Another safety measure that is considered is the PET metric, which is specifically better suited for intersecting conflicts. It is a temporal measure that uses the elapsed time between the departure of the first vehicle from a conflict area and the entry of the second vehicle to that area [12], [13]. To obtain the safety outputs, the SSM device available in SUMO is utilized. The SSM device output file of SUMO enables the extraction of information about conflicts such as the type of conflicts, the position of conflicts, etc. Conflicts are specified in accordance with established threshold values of TTC and PET metrics [56]. Although there is a current debate regarding the appropriate threshold values for TTC and PET measures for CAVs, widely adopted threshold values in the literature are employed in the context of this study. The TTC and PET metrics threshold values are adopted as 1.5 s and 5.0 s for HDVs, 0.75 s and 0.75 s for CAVs, respectively [12], [14], [29], [53], [70]. The designed scenarios are executed ten times using distinct seed values to replicate the stochastic nature of the traffic flow. The simulation step length is set as 0.1 s, and the simulation concludes when all vehicles have cleared the network in each scenario. Furthermore, an extra warm-up and cool-down period of 15 minutes is employed.

V. RESULTS AND DISCUSSION

The results section of this study is divided into three parts. Firstly, the simulation results comparing the baseline and mixed traffic scenarios for both networks are presented based on the selected evaluation measures, facilitating an assessment of the impact of increasing CAV penetration rates. Secondly, comprehensive evaluation scores are obtained by combining the findings of the microsimulation models and MCA methods. Finally, a sensitivity analysis is performed. The effects of changing the objective weighting method and employing different MCA methods on the findings of the study are investigated through a sensitivity analysis.

A. SIMULATION MODEL RESULTS

The outcomes of all scenarios derived from simulation models based on the selected evaluation measures are illustrated in Figure 7 and Figure 8. The best-performing scenario is bold-labeled, with the maximum value highlighted for the beneficial measure (C1) and the minimum values highlighted for the non-beneficial measures (C2, C3, C4, and C5).

Regarding the findings in terms of traffic efficiency, CAVs programmed with defensive driving behavior demonstrated poor performance, whereas CAVs with aggressive driving behavior showed superior performance compared to the base scenario. As the penetration rate of aggressive CAVs increased, there was a corresponding increase in average

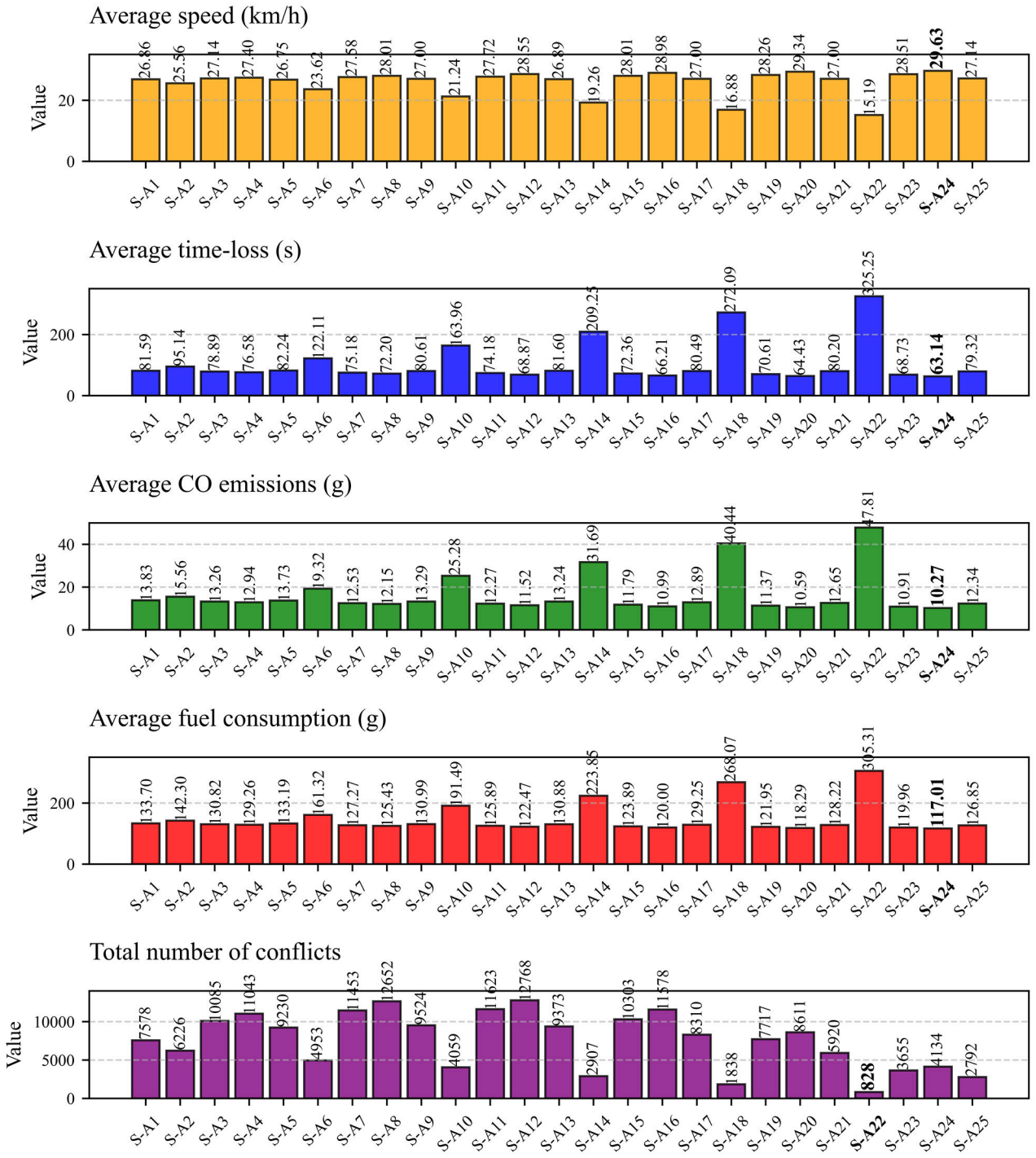


FIGURE 7. The simulation results of adopted scenarios for Network-A.

speed and a reduction in average time-loss across both networks. Conversely, a higher penetration rate of defensive CAVs resulted in a decrease in average speed and an increase in average time loss across both networks. Specifically, it was found that the average time-loss increased from 81.59 s (Scenario-A1) to 273.95 s (Scenario-A22) and from 46.84 s (Scenario-B1) to 182.29 s (Scenario-B22) with the 90% penetration rate of defensive-CAVs. On the other hand, it was observed that with the 90% penetration rate of CAVs with aggressive driving behavior, the average time-loss decreased

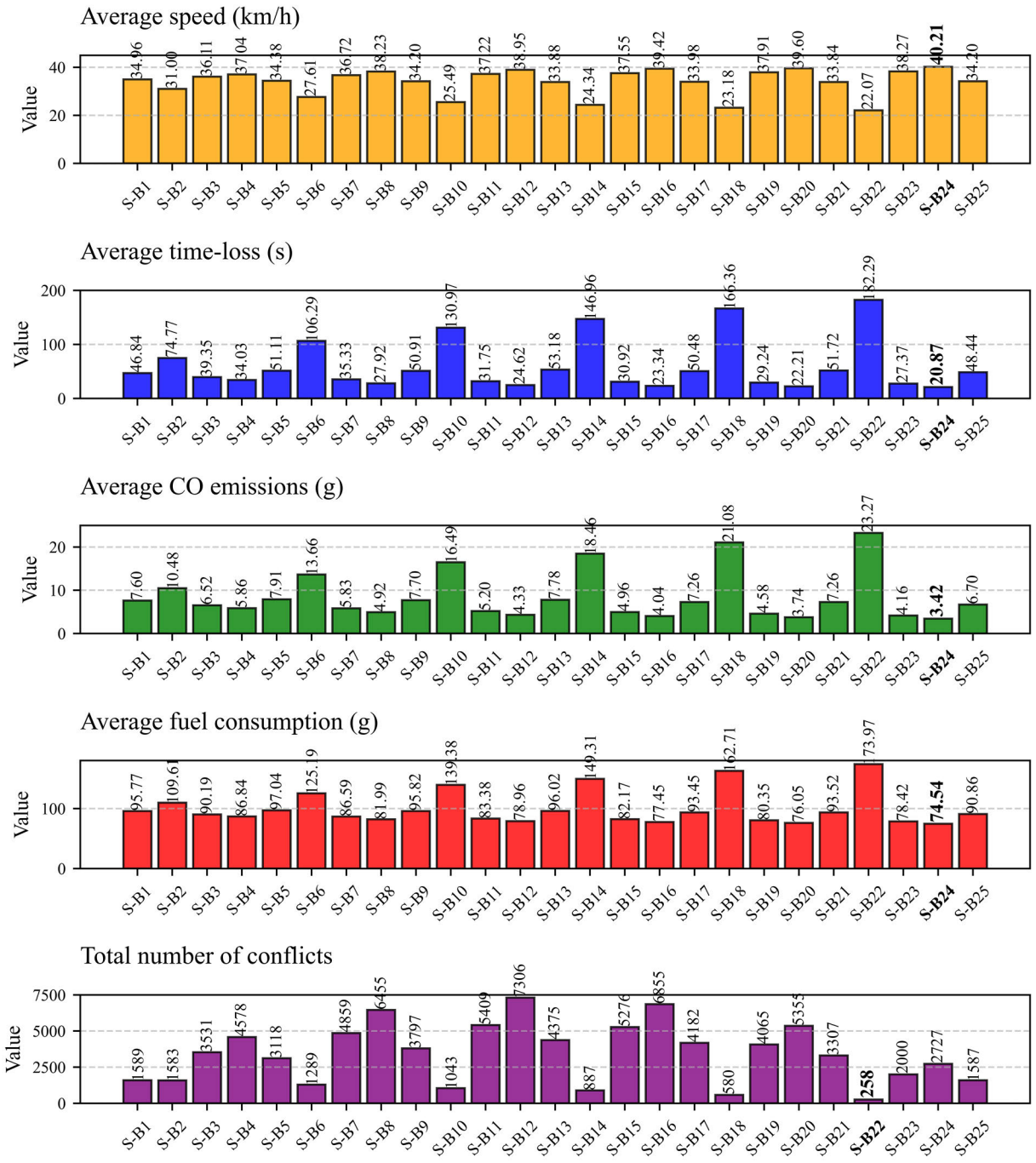


FIGURE 8. The simulation results of adopted scenarios for Network-B.

by approximately 22.6 % and 55.4 % compared with the base scenarios for Network-A and Network-B, respectively. As observed in Fig. 7 and Fig. 8, the defensive CAV scenarios (Scenario-A22 and Scenario-B22) lead to significantly higher time-loss compared to the aggressive CAV scenarios (Scenario-A24 and Scenario-B24). The decreased operational performance of defensive CAVs with regard to traffic efficiency can be attributed to the larger headway values maintained by defensive CAVs, which reduce traffic flow efficiency. The findings of our study on the traffic efficiency benefits of aggressive CAVs are consistent with the studies [11], [59], and [71], which found that aggressive driving behavior decreases average time-loss, while defensive driving behavior increases average time-loss. The scenarios

consisting of heterogeneous driving behaviors of CAVs and HDVs do not result in substantial improvements or deterioration in traffic efficiency across the examined penetration rates. This outcome can be attributed to the balanced distribution of different driving behaviors of CAVs, where the benefits and drawbacks of each behavior type neutralize one another.

As illustrated in Figure 7 and Figure 8, the increasing penetration rate of defensive CAVs contributes to an increase in CO emissions and fuel consumption. In contrast, aggressive CAVs are associated with a reduction in both CO emissions and fuel consumption. The penetration rates P1, P2, P3, P4, P5, and P6 of defensive CAVs resulted in significant increases in CO emissions and fuel consumption for Network-A compared to the base scenario. Specifically, CO emissions increased by approximately 12.5%, 39.7%, 82.8%, 129.1%, 192.4%, and 245.7%, while fuel consumption increased by 6.4%, 20.7%, 43.2%, 67.4%, 100.5%, and 128.4%, respectively. Similarly, for Network-B, the penetration rates P1, P2, P3, P4, P5, and P6 of defensive CAVs led to notable increases in CO emissions and fuel consumption compared to the base scenario. CO emissions increased by 37.9%, 79.7%, 117.0%, 142.9%, 177.4%, and 206.2%, while fuel consumption rose by 14.5%, 30.7%, 45.5%, 55.9%, 69.9%, and 81.7%, respectively. The corresponding penetration rates of aggressive CAVs resulted in decreases of approximately 6.4%, 12.1%, 16.7%, 20.5%, 23.4%, and 25.7% in CO emissions and 3.3%, 6.2%, 8.4%, 10.2%, 11.5%, and 12.5% in fuel consumption, respectively, for Network A, according to the base scenario. In Network-B, the decline rates were 22.9%, 35.3%, 43.0%, 46.8%, 50.8%, and 55.0% for CO emissions and 9.3%, 14.4%, 17.6%, 19.1%, 20.6%, and 22.2% for fuel consumption, corresponding to the penetration rates. Similar to the traffic efficiency results, scenarios consisting of heterogeneous driving behaviors of CAVs with HDVs did not bring about remarkable changes in CO emissions and fuel consumption compared with the base scenarios for both networks. As observed in Figure 7 and Figure 8, the study finds that while an increased penetration rate of defensive CAVs significantly raises CO emissions and fuel consumption, aggressive CAVs lead to reductions. The adverse impact of defensive CAVs was considered owing to the adoption of larger headway values and cautious performance, which aggravated the delay through networks, thus resulting in increased emissions and fuel consumption values. The findings of our study on environmental performance are consistent with the study [59], which found that defensive driving behavior leads to decreases in network efficacy, prolongs the duration to traverse the network, and emits more emissions.

As depicted in Figure 7 and Figure 8, regarding the findings in terms of traffic safety, defensive CAVs outperformed normal and aggressive CAVs. The total number of conflicts decreased by 17.8%, 34.6%, 46.4%, 61.6%, 75.7%, and 89.1% at P1, P2, P3, P4, P5, and P6 penetration levels of defensive CAVs in Network-A, and by 0.4%, 18.9%, 34.4%, 44.2%, 63.5%, and 83.8% in Network-B, respectively, according to the base scenarios. In contrast to the performance of defensive CAVs, the total number of conflicts in Network-A increased compared to the baseline scenario up to a 45% penetration of aggressive CAVs. Beyond this penetration level, a declining trend in the number of conflicts was observed. At the 90% penetration rate of aggressive CAVs, the total number of conflicts in Network-A decreased by 45.4% compared to the base scenario. Accordingly, in Network-B, the total number of conflicts peaked at a 45% penetration rate of aggressive CAVs. Beyond this threshold, a downward trend in the total number of conflicts was observed. The safety benefits of scenarios consisting of heterogeneous driving behaviors of CAVs are observed with increasing penetration rates. The superior performance of defensive CAVs can be linked to their adoption of larger headway values compared to normal and aggressive driving behaviors. The traffic safety findings of our study align with those of previous studies [56], [72], which observed that CAVs adopting shorter headway values tend to increase the number of conflicts, particularly at low penetration levels at signalized intersections. In contrast to the findings of study [11], which reported safety improvements with higher CAV penetration rates, our study found that safety improvements are more pronounced with defensive CAVs, suggesting that driving behavior significantly influences safety outcomes.

B. MULTI-CRITERIA ANALYSIS RESULTS

As mentioned earlier, to establish a comprehensive evaluation of the impacts of different driving behaviors of CAVs on urban networks, each adopted scenario is evaluated using selected measures in terms of traffic efficiency, environmental performance, and traffic safety. The weights of the criteria are determined using the CRITIC method, and the CoCoSo method is employed to calculate the final scores of scenarios. For each penetration rate, five scenarios are designed and evaluated, including the penetration of defensive, normal, and aggressive CAVs, a heterogeneous CAV fleet, and the baseline scenario. The baseline scenario is included at each penetration rate to serve as a reference point, enabling a systematic assessment of performance variations as CAV penetration increases. The findings of the simulation models are used to create decision matrices as the first step in the multi-criteria analysis process. The procedure using Eqs. (1)-(13) is applied to calculate the weights of the criteria and final evaluation scores for each considered penetration rate and network, separately. The sum of the weights should be equal to one, and the calculated criteria weights using the CRITIC method are listed in Table 7. According to the results of the CRITIC method presented in Table 7, the most significant evaluation criterion is C5, which is highlighted in bold and relates to traffic safety for each penetration rate. A larger standard deviation signifies more pronounced variation in the evaluation measures, thereby yielding a greater amount of

information and stronger evaluation intensity. Consequently, a higher weight is assigned to the criterion related to traffic safety (C5) in the CRITIC method. Furthermore, it is found that the weights of C1, C2, C3, and C4 are similar for each penetration rate. This means that the level of intra-criteria contrast is low, whereas the correlation between these criteria is strong.

Figure 9 depicts the final scores of the scenarios for each penetration rate in Network-A, with the best-performing scenarios bolded. As can be seen from the figure, the base scenario, comprising only HDVs (Scenario-A1), outperformed all other scenarios up to a 60% penetration rate. Beyond this threshold, scenarios consisting of heterogeneous driving behaviors of CAVs combined with HDVs (Scenario-A21 and Scenario-A25) achieved the highest rankings.

This outcome is attributed to the fact that the safety benefits of heterogeneous CAV fleets become more pronounced beyond the 60% penetration level as compared to the base scenario in Network-A. On the other hand, the scenarios that exhibited the worst performance were those consisting of defensive CAVs combined with HDVs (Scenario-A2, Scenario-A6, Scenario-A10, Scenario-A14, and Scenario-A22) in Network-A.

TABLE 7. The calculated criteria weights using CRITIC method.

Network Type	P. Rate	Criteria Weights				
		C1	C2	C3	C4	C5
Network-A	P1 (15%)	0.1216	0.1222	0.1239	0.1241	**0.5082**
	P2 (30%)	0.1243	0.1260	0.1270	0.1271	**0.4956**
	P3 (45%)	0.1248	0.1279	0.1282	0.1285	**0.4906**
	P4 (60%)	0.1273	0.1312	0.1313	0.1315	**0.4787**
	P5 (75%)	0.1249	0.1306	0.1299	0.1301	**0.4845**
	P6 (90%)	0.1278	0.1385	0.1363	0.1369	**0.4605**
Network-B	P1 (15%)	0.1165	0.1168	0.1184	0.1189	**0.5294**
	P2 (30%)	0.1188	0.1195	0.1206	0.1210	**0.5201**
	P3 (45%)	0.1203	0.1213	0.1224	0.1227	**0.5133**
	P4 (60%)	0.1212	0.1220	0.1231	0.1234	**0.5102**
	P5 (75%)	0.1249	0.1257	0.1266	0.1267	**0.4961**
	P6 (90%)	0.1306	0.1335	0.1330	0.1330	**0.4699**

Figure 10 depicts the final scores of the scenarios for each penetration rate in Network-B, with the best-performing scenarios bolded. As illustrated in Figure 10, the base scenario, consisting solely of HDVs (Scenario-B1), outperformed all other scenarios up to a 75% penetration rate. Beyond this level, the scenario consisting of heterogeneous driving behaviors of CAVs combined with HDVs (Scenario-B25) attained the highest ranking. The improvements associated with heterogeneous CAV fleets become more pronounced beyond the 75% penetration rate when compared to the base scenario in Network-B. Consistent with the findings in Network-A, the scenarios that demonstrated the worst performance were those involving defensive CAVs combined with HDVs (Scenario-B2, Scenario-B6, Scenario-B10, Scenario-B14, and Scenario-B22) in Network-B.

C. SENSITIVITY ANALYSIS

The use of criteria weighting methods and the selection of different MCA methods can influence the outcomes [73]. Therefore, the effects of changing the objective weighting method and employing different MCA methods on the findings of the study are investigated in this part. Firstly, the Entropy method is utilized to compare the criteria weights derived from the CRITIC method and to evaluate the scenarios with the CoCoSo method. The primary reason for selecting the Entropy method in this study is its widespread application within the field of transportation engineering [37], [38], [39].

The Entropy method is a mathematical approach utilized to assign weights to criteria based on the variability and distribution of data. The fundamental principle of the Entropy method is that greater variation in the values of a variable results in lower entropy, while less variation leads to higher entropy. In information theory, entropy serves as a model for quantifying the level of uncertainty represented by a discrete probability distribution [74]. The calculation steps of the Entropy method can be found in study [75]. Table 8 presents the criteria weights computed using the Entropy method at each penetration level, with the highest weights bolded.

TABLE 8. The calculated criteria weights using entropy method.

Network Type	P. Rate	Criteria Weights				
		C1	C2	C3	C4	C5
Network-A	P1 (15%)	0.0110	0.1149	0.0817	0.0220	**0.7704**
	P2 (30%)	0.0195	0.2292	0.1759	0.0501	**0.5253**
	P3 (45%)	0.0260	0.3158	0.2538	0.0787	**0.3257**
	P4 (60%)	0.0284	**0.3503**	0.2920	0.0965	0.2328
	P5 (75%)	0.0316	**0.3696**	0.3156	0.1132	0.1700
	P6 (90%)	0.0304	**0.3382**	0.2949	0.1116	0.2249
Network-B	P1 (15%)	0.0122	0.2619	0.1395	0.0221	**0.5643**
	P2 (30%)	0.0163	0.3289	0.1875	0.0319	**0.4354**
	P3 (45%)	0.0183	0.3533	0.2251	0.0411	**0.3622**
	P4 (60%)	0.0196	**0.3693**	0.2530	0.0482	0.3099
	P5 (75%)	0.0206	**0.3833**	0.2862	0.0578	0.2521
	P6 (90%)	0.0226	**0.4050**	0.3229	0.0671	0.1824

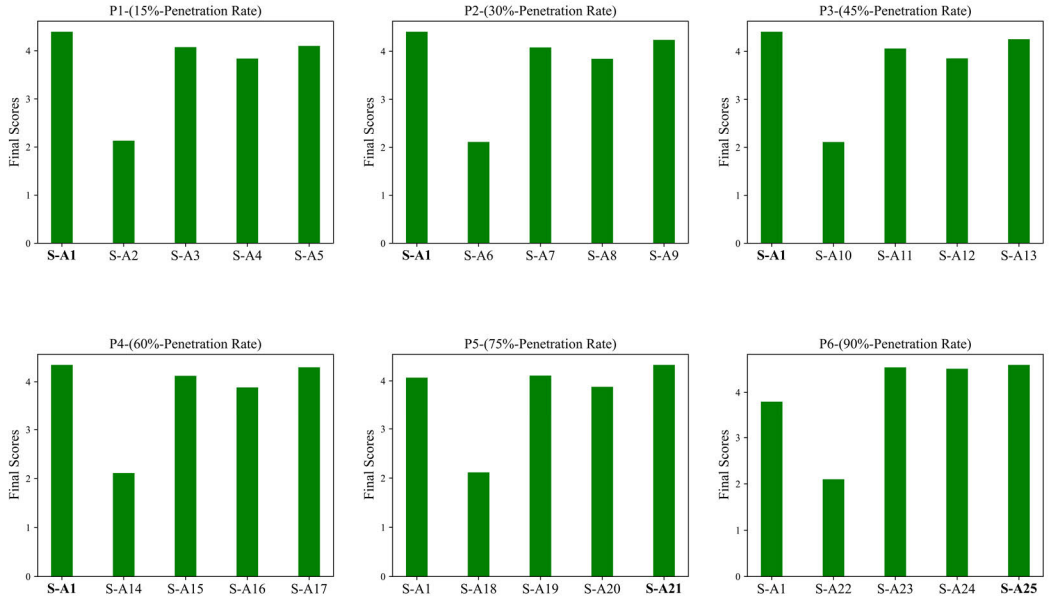


FIGURE 9. The final scores of adopted scenarios in Network-A.

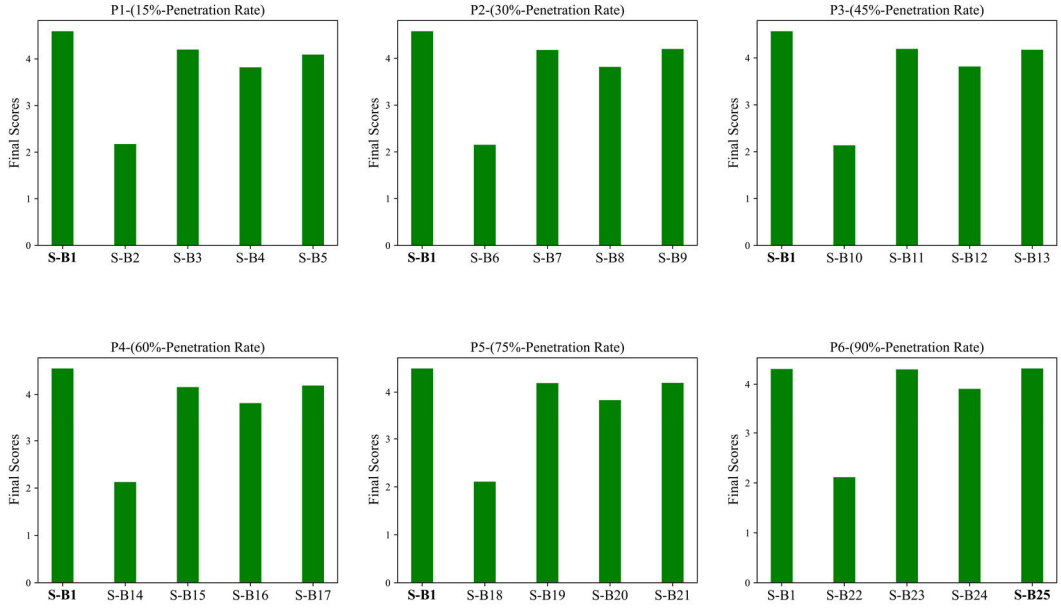


FIGURE 10. The final scores of adopted scenarios in Network-B.

Based on the results of the Entropy method shown in Table 8, criterion C5 which is related to traffic safety emerges as the most significant evaluation criterion for the P1, P2, and P3 penetration rates in both networks. For the P4, P5, and P6 penetration rates, the highest weight shifts to criterion C2, which represents average time-loss. While the CRITIC and Entropy methods give the highest weight to the same criterion at P1, P2, and P3 penetration rates, the results differ at P4, P5, and P6 penetration rates. The divergence in results at P4, P5, and P6 can be attributed to differences in how these methods evaluate criteria. The CRITIC method emphasizes the variability (standard deviation) and contrast of each criterion while also accounting for correlations between criteria. Conversely, the Entropy method focuses solely on the distribution and dispersion of data within each criterion, without considering inter-criteria relationships. At P4, P5, and P6 penetration rates, the distribution of average time loss (C2) exhibits greater variability according to the base scenario,

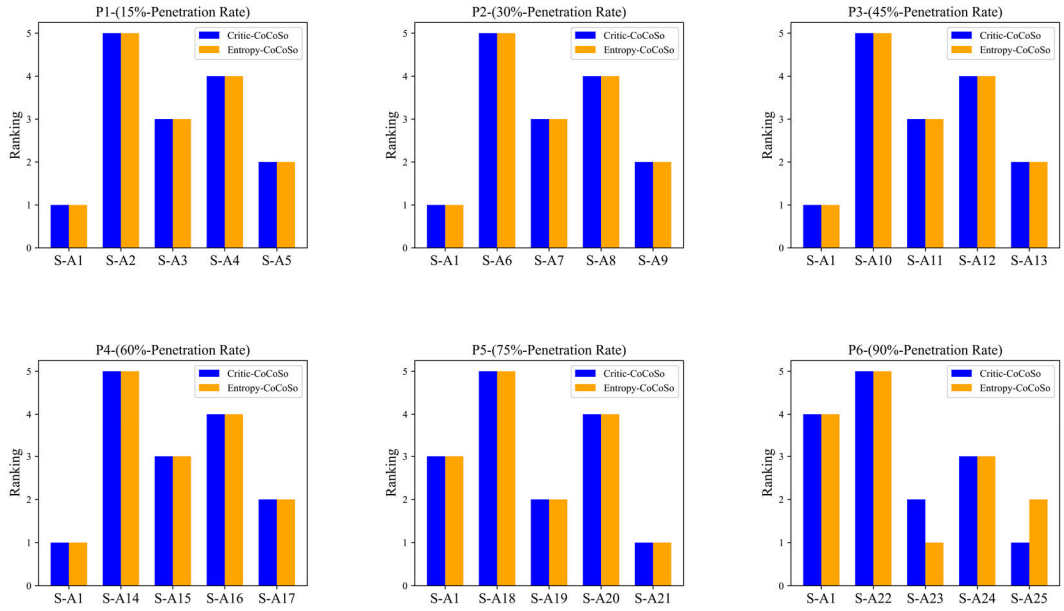


FIGURE 11. The comparison of using different weighting methods on scenario rankings for Network-A.

leading Entropy to assign it a higher weight. The comparison of scenario rankings obtained using the CRITIC and Entropy weighting methods, in conjunction with the CoCoSo method, is illustrated in Figure 11 for Network-A. It can be observed from the Figure 11, the CoCoSo method with different weighting methods have same ranking patterns at P1, P2, P3, P4 and P5 penetration rates for Network-A. However, a slight discrepancy is noted at the P6 penetration rate. Specifically, while the CRITIC-CoCoSo method assigns the highest ranking to the Scenario-A25, the Entropy-CoCoSo method ranks Scenario-A23 as the highest at this penetration rate.

Furthermore, the comparison of scenario rankings obtained using the CRITIC and Entropy weighting methods, combined with the CoCoSo method, is depicted in Figure 12 for Network-B. Although slight differences are observed in the intermediate rankings, the highest and lowest rankings remain consistent at the P1, P2, P3, P4, and P5 penetration rates. Similar to the findings of Network-A, the highest ranking changed at the P6 penetration rate. While the CRITIC-CoCoSo method assigns the highest ranking to Scenario-B25, the Entropy-CoCoSo method ranks Scenario-B23 as the highest at this penetration rate. From Figure 11 and Figure 12, it can be concluded that changing the weighting method has a slight impact on the rankings of scenarios. The highest and lowest rankings demonstrate consistency across most scenarios, indicating robustness in the evaluation outcomes despite variations in weighting approaches.

As the second step of the sensitivity analysis, the results obtained by the CRITIC-CoCoSo method are compared with those obtained using different MCA methods, specifically the Preference Ranking Organization Method for Enrichment of Evaluations II (PROMETHEE II) and the Multi-Attributive Border Approximation Area Comparison (MABAC). These methods are chosen for their capability to handle multiple conflicting criteria, provide robust evaluations of scenarios, and for their established application in addressing transportation engineering problems in the literature [76], [77], [78], [79]. The detailed calculation steps of both methods can be found in studies [79], [80].

The findings of the study obtained by the CoCoSo method in conjunction with the CRITIC weighting method are compared with those derived from different MCA methods, including PROMETHEE II and MABAC. This comparison aims to assess the consistency of scenario rankings across these diverse evaluation approaches. The comparison of scenario rankings derived from different MCA methods at each penetration rate for Network-A is presented in Figure 13. As depicted in Figure 13, all methods demonstrate consistency in assigning the first rank across all penetration rates. The base scenario, consisting of only HDVs (Scenario-A1), demonstrated superior performance compared to all other scenarios at penetration rates of up to 60%. Beyond this threshold, scenarios consisting of heterogeneous driving behaviors of CAVs in combination with HDVs (Scenario-A21, and Scenario-A25) emerged as the highest-ranked alternatives for all methods. It is observed that the baseline scenario loses its dominant position after a 60% penetration rate and subsequently drops to lower ranks as the penetration rate of CAVs increases. From the comprehensive evaluation results, it can be concluded that the scenario comprising only HDVs performs optimally at low penetration rates but becomes progressively less effective as the CAV penetration rate increases for Network-A. The worst

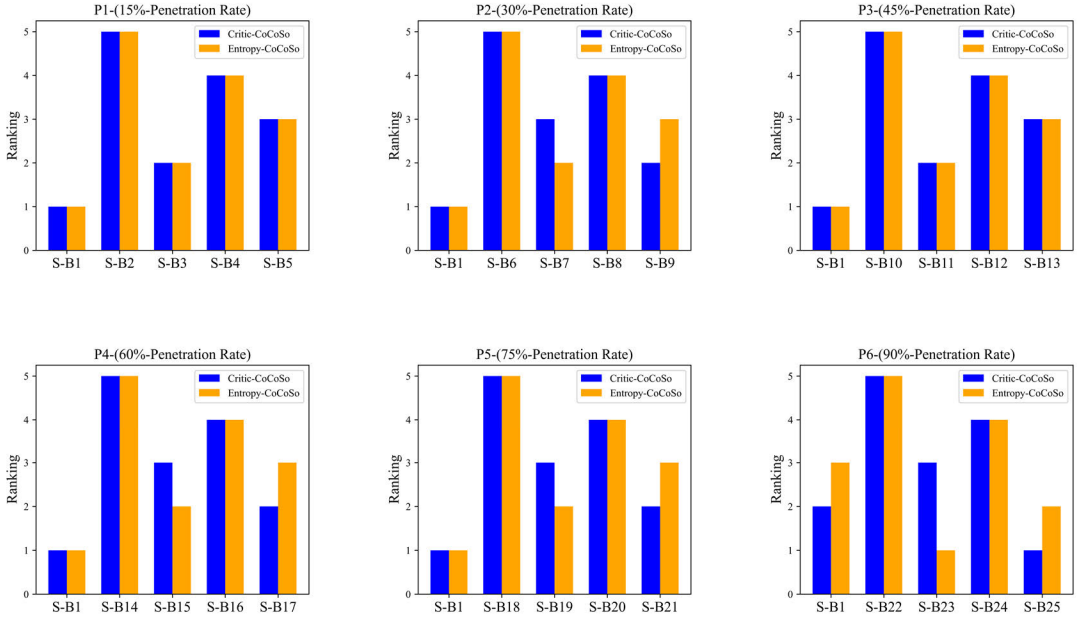


FIGURE 12. The comparison of using different weighting methods on scenario rankings for Network-B.

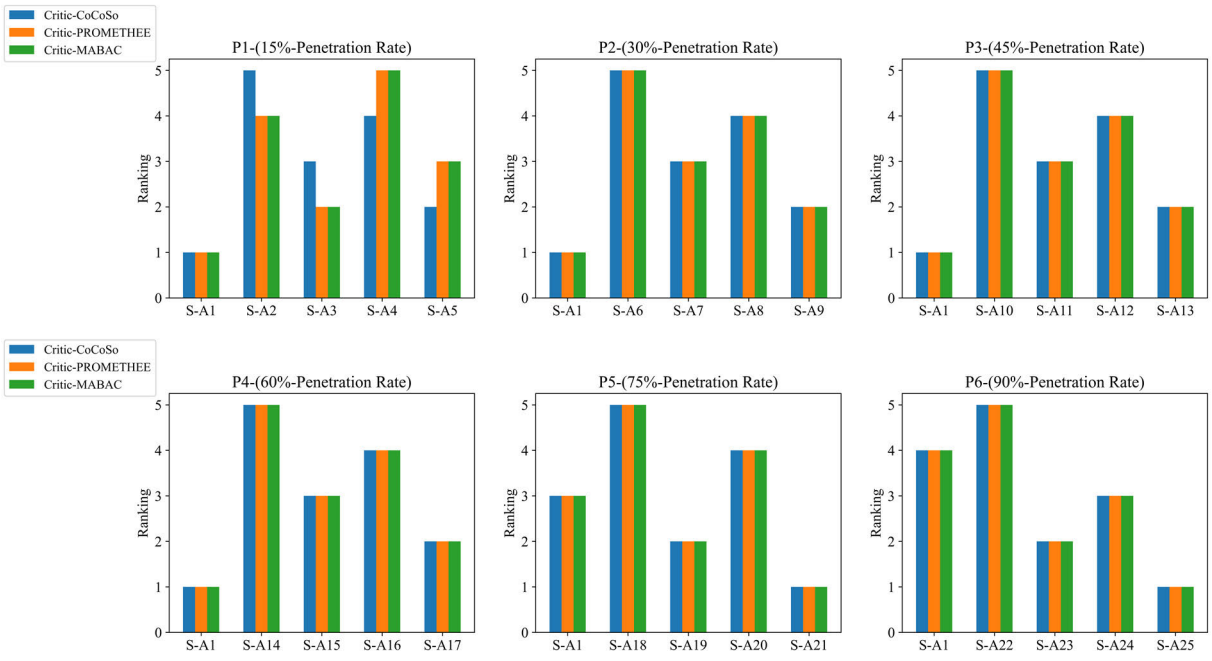


FIGURE 13. The comparison of scenario rankings using different MCA methods for Network-A.

performance is mostly obtained by scenarios consisting of defensive CAVs and HDVs for all methods, especially highlighting its limitations at higher penetration rates. Scenarios with heterogeneous driving behaviors of CAVs with HDVs demonstrate improved performance as the penetration rate increases. Furthermore, as shown in Figure 13, the rankings provided by the CoCoSo, PROMETHEE II, and MABAC methods exhibit consistency across most scenarios for Network-A.

The comparison of scenario rankings obtained from different MCA methods at each penetration rate for Network-B is illustrated in Figure 14. As shown in Figure 14, all methods consistently assigned the first rank to the scenario consisting exclusively of HDVs (Scenario-B1) at penetration rates of up

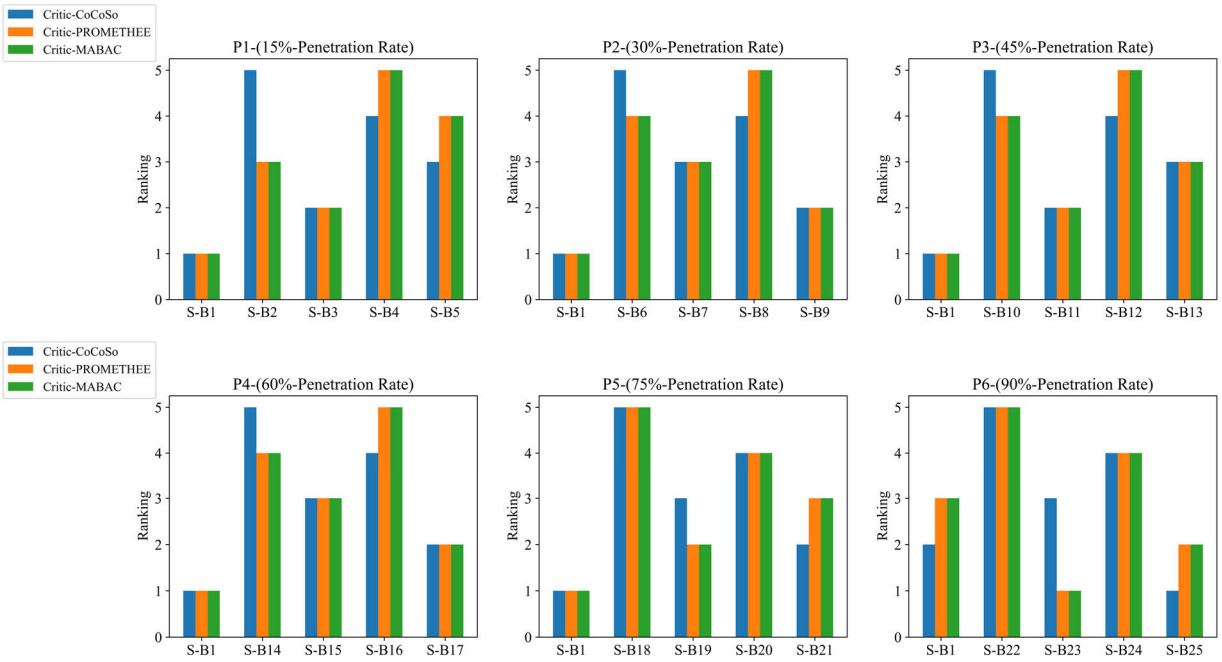


FIGURE 14. The comparison of scenario rankings using different MCA methods for Network-B.

to 75%. Beyond this threshold, the baseline scenario loses its dominant position across all methods. However, a slight variation is observed regarding the first-ranked scenario at the 90% penetration rate. The CoCoSo method assigns the first rank to Scenario-B25, whereas the PROMETHEE II and MABAC methods rank Scenario-B23 as the top scenario. On the other hand, the scenario with the worst performance varies between the methods at the P1, P2, P3, and P4 penetration rates. While the CoCoSo method identifies the defensive CAVs scenarios as having the worst performance, the PROMETHEE II and MABAC methods assign the worst performance to the aggressive CAVs scenarios. At the P5 and P6 penetration rates, all methods consistently identify the defensive CAVs scenarios as having the worst performance, highlighting its limitations at higher penetration rates. The divergence in scenario rankings, particularly at lower penetration rates, is more pronounced for Network-B compared to Network-A. This can be attributed to the more complex network configurations and traffic dynamics of Network-B. Additionally, the slight differences between the methods may stem from the varying mathematical processing approaches employed by each method, which influence how criteria and rankings are evaluated.

The CoCoSo method emphasizes identifying compromise solutions by balancing multiple criteria, the PROMETHEE II method focuses on pairwise comparisons to establish preference rankings, and the MABAC method evaluates alternatives based on their distance from the border approximation area. It can be concluded that, although some variations in scenario rankings are observed for both networks, the first-ranked scenario is consistently identified across the methods in most cases, indicating a high level of agreement in determining the top-performing scenario.

VI. CONCLUSION

This study proposes a framework based on combining multi-criteria analysis methods and traffic microsimulation modeling to establish a comprehensive evaluation of the impacts of mixed traffic conditions on urban networks. While SUMO is employed as a microscopic traffic simulator, the CRITIC-CoCoSo methods are employed to comprehensively evaluate the scenarios. Two real urban networks are modeled using SUMO in the context of this study. Six penetration rates, namely P1 (15%), P2 (30%), P3 (45%), P4 (60%), P5 (75%), and P6 (90%), and three different driving behaviors of CAVs, namely defensive, normal, and aggressive, are taken into consideration in the scope of this study. The average speed, average time-loss, average CO emission and fuel consumption, and the total number of conflicts are chosen as evaluation measures in terms of traffic efficiency, environmental performance, and traffic safety, respectively.

Based on the results regarding traffic efficiency, it is found that while defensive CAV behavior leads to an increase in the average time-loss and a decrease in average speed, the penetration of aggressive CAV behavior reduces the average time-loss and increases the average speed compared to the base scenarios for both networks. These impacts become increasingly pronounced as the penetration rate of CAVs grows. Similar to the findings on traffic efficiency, defensive CAV behavior resulted in an increase in CO emission and fuel consumption, whereas aggressive CAV behavior contributed to a decrease. Despite their higher acceleration

capabilities, the emission mitigation observed in aggressive CAVs can be linked to reduced travel times and acceleration rates throughout their trips, thereby promoting a smoother driving experience compared to HDVs. In terms of traffic safety, defensive CAV behavior is identified as the safest driving behavior among the other driving behaviors of CAVs.

According to the results of the applied multi-criteria analysis methodology, the most significant criterion is found to be C5 - "Total number of conflicts" for both networks at each penetration rate. Based on the comprehensive evaluation results, the base scenario, comprising only HDVs, demonstrated superior performance compared to all other scenarios up to a 60% penetration rate in Network-A. However, beyond this threshold, scenarios featuring heterogeneous driving behaviors of CAVs combined with HDVs emerged as the top-performing alternatives. On the other hand, the base scenario, consisting solely of HDVs, outperformed all other scenarios up to a 75% penetration rate for Network-B. Beyond this level, the scenario consisting of heterogeneous driving behaviors of CAVs combined with HDVs attained the highest ranking for Network-B. Furthermore, the effects of changing the objective weighting method and employing different MCA methods on the findings of the study are investigated within the scope of this study. Firstly, the Entropy method is employed to compare the criteria weights obtained from the CRITIC method and to evaluate the scenarios using the CoCoSo method. Later, the findings of the study, obtained using the CoCoSo method combined with the CRITIC weighting method, are compared with those derived from other MCA methods, including PROMETHEE II and MABAC. Despite the changing of the weighting method and the application of different MCA methods, the identification of the first-ranked scenario demonstrates consistency across the methods in most cases. As a summary of the comprehensive evaluation, while aggressive CAV behavior has the potential to enhance traffic efficiency and reduce harmful environmental impacts, particular attention should be given to the interaction between HDVs and aggressive CAV behavior in mixed traffic conditions. While aggressive CAVs contribute to improved traffic efficiency, they also introduce safety risks, especially at lower penetration rates. In this context, ethical risks and safety implications associated with aggressive driving behavior should be taken into account. Conversely, defensive CAVs enhance traffic safety, though at the cost of reduced traffic efficiency. The findings of this research could potentially be incorporated to prompt regulatory bodies to adopt effective strategies concerning CAV operation, aiming to enhance traffic efficiency, improve traffic safety, and reduce traffic emissions. The outcomes of this study suggest that policymakers should consider the trade-offs between traffic efficiency and traffic safety when promoting CAV deployment.

Although this study presents noteworthy findings, it has some limitations and key issues that need to be addressed. It is assumed that the driving behavior of HDVs will not change in the presence of different driving behaviors of CAVs. Moreover, validating the model parameters for CAV driving behaviors with real-world data is essential. Another key point to note is that the default car-following model of SUMO, namely the Krauss car-following model, is employed for modeling the longitudinal behavior of HDVs, whereas the IDM model is used for CAVs. Since each car-following model relies on specific assumptions, as investigated in the study [81], future research could utilize different car-following models to represent the behaviors of HDVs and CAVs. Six penetration rates are integrated into this study. In mixed traffic condition scenarios, one type of driving behavior is employed to reveal the exact impact of different driving behaviors of CAVs on selected evaluation measures. Additionally, the scenario that consists of an equal percentage of each driving behavior of CAVs at each penetration rate is employed to reflect a more realistic traffic condition. It is emphasized that an equal percentage of each driving behavior is assumed in this scenario type. Future studies could explore different combinations of penetration rates of different driving behaviors of CAVs. The proposed methodology could be extended to larger urban networks or applied to other road environments, such as freeways. Finally, future studies should focus on developing adaptive driving behaviors in CAVs to optimize traffic efficiency, traffic safety, and environmental impacts, considering varying traffic demands.

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